



**Molinaroli College of
Engineering and Computing**

**Thermal Analysis and Optimization of a Passive Finned
Heatsink for a High-Performance Microchip**

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Motivation

Microchips are central to modern computers, servers, and mobile devices because they process and transmit information at high speed within a compact footprint. As their power density increases, localized hot spots become more likely, which can reduce performance and damage sensitive components. Effective thermal management is therefore necessary to maintain reliability and prevent thermal throttling.

Conventional cooling methods such as fans and liquid coolants can remove heat effectively, but they also introduce disadvantages including added power consumption, mechanical wear, and potential leakage. A passive heatsink offers a simpler alternative by using natural convection and finned surfaces to remove heat without moving parts. The goal of this project was to determine whether a passive finned heatsink could provide sufficient cooling for a high-performance microchip under realistic geometric constraints.

Background

Heat in a microchip is generated when electrical resistance within the transistor network converts part of the electrical energy into thermal energy. Although each transistor is extremely small, the large number of transistors packed into a modern chip allows this heat to accumulate quickly. If the temperature rises too far, the chip may reduce its performance through thermal throttling or experience permanent material damage, particularly at solder joints and other thermally sensitive interfaces.

Natural convection provides a passive method of cooling by allowing hot air near the chip surface to rise and be replaced by cooler surrounding air. On its own, however, natural convection often cannot remove enough heat from a high-power device. For this reason, passive cooling systems commonly use fins to increase the available heat-transfer area. Previous studies have shown that the performance of naturally cooled fin arrays depends strongly on fin spacing, fin geometry, and surface properties. Vertical plate-fin systems exhibit an optimum spacing under natural convection, and both geometry and emissivity can significantly affect thermal resistance and heat dissipation.¹⁻⁴ These findings support the present study's focus on overall heatsink height, fin spacing, and emissivity as key design variables.

Model Introduction

The heatsink in this study was modeled in AutoCAD as a passive straight-fin cooling device for a high-performance microchip. The model was intended to capture the geometric features most important to passive heat removal while remaining simple enough for analytical thermal evaluation. As shown in Figures 1 through 3, the final design consists of a solid base and a parallel array of vertical fins that create open channels for buoyancy-driven airflow. The geometric envelope was selected to remain compatible with a server-class Intel platform based on Clearwater Forest and a Supermicro X14SBH / SYS-212H-TN system family associated with the Socket E2/LGA-4710 form factor.⁵⁻⁹ The final modeled dimensions were approximately 95.0 mm × 70.5 mm × 85.3 mm. Copper was chosen as the heatsink material because its high thermal conductivity improves heat spreading from the chip contact region into the fin array. This geometry provided the basis for the thermal model and the subsequent parametric analysis.

Methods

A passive straight-fin heatsink was modeled to determine whether a high-performance microchip could be cooled by natural convection and radiation alone. The design target was based on a maximum junction temperature of 85 °C, an ambient temperature of 20 °C, and a required heat dissipation of 350 W. The allowable temperature rise and corresponding thermal design criterion were defined by:

$$R_{\theta, total} \leq \frac{T_{j, max} - T_{\infty}}{Q}$$

where $R_{\theta, total}$ is the maximum allowable junction-to-ambient thermal resistance. Detailed substitutions are provided in Appendix A.

The heatsink was constrained to a footprint of 95 mm by 70.5 mm with a base thickness of 3.0 mm. Within this envelope, the design variables were fin height, fin spacing, surface emissivity, and effective chip-to-base contact area. Heat flow from the microchip to the surroundings was represented as a series resistance network:

$$R_{\theta, total} = R_{\theta, TIM} + R_{\theta, base} + R_{\theta, sink-air}$$

where the three terms represent the thermal interface resistance, base conduction resistance, and sink-to-air resistance, respectively. Base conduction was treated as one-dimensional through-thickness conduction, and the contact area was modeled as a fraction of the package area because the exact processor heat-spreader geometry was not publicly available.

The heatsink footprint was selected to remain compatible with the assumed server-class board and chassis environment rather than representing an unconstrained standalone heatsink. This allowed the parametric study to be performed within a hardware-relevant geometric envelope. Heat rejection from the finned surfaces was modeled as the combined effect of natural convection and radiation. The fins were approximated as vertical surfaces, and buoyancy-driven flow was characterized through the Rayleigh number:

$$Ra_H = \frac{g\beta\Delta TH^3}{\nu\alpha}$$

followed by a vertical-surface Nusselt correlation to estimate the convection coefficient. Radiation was incorporated through a linearized radiative coefficient, and the convective and radiative contributions were combined into an effective air-side coefficient used to calculate the sink-to-air thermal resistance.

A parametric study was performed by varying overall heatsink height and fin spacing, while emissivity and effective contact area were treated as secondary sensitivity variables. For each case, the total exposed area, effective heat-transfer coefficient, total thermal resistance, and maximum dissipatable heat rate were calculated according to:

$$Q_{max} = \frac{T_{j,max} - T_{\infty}}{R_{\theta,total}}$$

The final design was selected as the smallest passive geometry within the allowable footprint that satisfied the temperature constraint at the target heat load. The analysis further assumed constant solid properties, representative air properties evaluated at the film temperature, and steady-state passive cooling without forced airflow.

Results

The final design selected from the parametric study was a straight-fin copper heatsink with a footprint of $95 \text{ mm} \times 70.5 \text{ mm}$ and an overall height of 85.3 mm. With a base thickness of 3.0 mm, the usable fin height was 82.3 mm. The final array contained 24 fins, each 1.0 mm thick with 2.0 mm spacing, and the surface emissivity was assumed to be 0.95.

As shown in Figure 4, the predicted maximum heat dissipation increased with overall heatsink height. Taller fins increased the exposed area available for natural convection and radiation, which raised Q_{max} . The dashed line marks the 350 W design target, showing that the required heat load was met only near the upper end of the allowable height range.

The effect of fin count is shown in Figure 5. As the number of fins increased, the fin spacing decreased, which increased the total exposed area and improved the predicted heat dissipation. This confirms that passive cooling performance was sensitive to fin spacing and fin count over the selected geometric range.

The final sink-to-air thermal resistance was 0.1795 K/W, and the corresponding total thermal resistance was 0.18427 K/W. The predicted maximum heat dissipation was 352.7 W, which slightly exceeded the 350 W target. This indicates that the selected geometry could meet the thermal requirement within the assumptions of the passive model. The analysis also showed that air-side resistance dominated the system, while the interface and base-conduction terms were comparatively small.

Limitations

This model was developed as a preliminary analytical design tool and includes several simplifying assumptions. The parametric studies were performed one variable at a time, which isolates the effect of each parameter but does not capture interactions between fin geometry, fin count, airflow behavior, and contact conditions. The fin-height study mainly reflects the increase in exposed area with taller fins, while the fin-count study reflects the tradeoff between available surface area and the fixed heatsink width.

Additional limitations arise from the use of a resistance-network framework instead of a full three-dimensional CFD model. The chip-to-heatsink contact area was estimated as a fraction of the package area, the thermal interface was represented with a single thickness and conductivity, and the natural-convection coefficient was obtained from standard correlations using representative air properties. Solid properties were treated as constant, and air properties were evaluated at a representative film temperature rather than allowed to vary spatially throughout the fin channels.

. Radiation was included through a linearized coefficient and a single assumed emissivity value. These assumptions are appropriate for a preliminary design study, but they neglect detailed temperature nonuniformity, local airflow interactions, enclosure effects, and transient behavior. With additional computational resources and time, these limitations could be addressed using a three-dimensional CFD or conjugate heat-transfer model with variable fluid properties, more detailed package geometry, and improved resolution of enclosure and airflow effects.

With additional computational resources and time, these limitations could be addressed using a three-dimensional CFD or conjugate heat-transfer model with variable fluid properties, more detailed package geometry, and improved resolution of enclosure and airflow effects.

Conclusions

The results show that passive microchip cooling performance depends strongly on fin geometry, surface properties, and material selection. The final design was a straight-fin copper heatsink with an overall height of 85.3 mm, a base thickness of 3.0 mm, and a 24-fin array. Under the assumed contact and emissivity conditions, the design achieved a predicted maximum heat dissipation of 352.7 W, slightly exceeding the 350 W target.

The analysis also showed that the dominant thermal limitation was the sink-to-air resistance rather than conduction through the base or interface. As a result, overall height, fin spacing, and emissivity were the most important design variables in the passive system. Copper was retained as the final material because its high thermal conductivity improved heat spreading and reduced internal resistance.

Overall, the study demonstrates that a passive finned heatsink can satisfy the target thermal load under favorable conditions, but the performance margin is small. The final design exceeded the 350 W target by only 2.7 W, indicating that the passive solution was feasible but operated close to its practical thermal limit. Future work should focus on validating the final geometry experimentally or with a more detailed three-dimensional simulation before applying the design to practical server hardware.

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Sample Calculations

The design target was based on a conservative maximum junction temperature of $T_{j,max} = 85^{\circ}\text{C}$ with an ambient temperature of $T_{\infty} = 20^{\circ}\text{C}$. So, the total allowable junction-to-ambient temperature rise was:

$$T_{j-a} = T_{j,max} - T_{\infty} \quad (1)$$

$$T_{j-a} = 85 - 20 = 65^{\circ}\text{C}$$

The thermal design criterion was then expressed by:

$$Q_{max} = \frac{T_{j,max} - T_{\infty}}{R_{\theta,total}} \quad (2)$$

This expression can be rearranged to follow the equivalence:

$$R_{\theta,total} \leq \frac{T_{j,max} - T_{\infty}}{Q} \quad (2.1)$$

For a target heat dissipation of approximately 350 W, the required overall thermal resistance is:

$$R_{\theta,total,target} = \frac{65}{350} = 0.186 \text{ K/W}$$

The overall resistance can be decomposed into three terms:

$$R_{\theta,total} = R_{\theta,TIM} + R_{\theta,base} + R_{\theta,sink-air} \quad (3)$$

In this equation, $R_{\theta,TIM}$ is the thermal interface resistance between chip and heatsink, $R_{\theta,base}$ is the conduction resistance through the heatsink base, and $R_{\theta,sink-air}$ is the resistance associated with passive heat rejection from the heatsink to the surrounding air.

The heatsink was constrained to fit inside of the server rack. The final footprint used was $W = 70.5 \text{ mm}$, $L = 95 \text{ mm}$, and $t_b = 3.0 \text{ mm}$. These values were chosen as a practical compromise. The base is thick enough to provide mechanical stiffness and laterally spread heat while remaining thin enough that the conduction penalty through the base is small enough compared to air-side resistance.

The overall heatsink height selected was $H = 85.3 \text{ mm}$. This gives a usable fin height of:

$$H = H_{total} - t_b \quad (4)$$

$$H = 85.3 - 3.0 = 82.3 \text{ mm}$$

The heatsink was modeled as a straight-fin array with fins aligned in parallel. The final fin geometry was $t_f = 1.0 \text{ mm}$ and $s = 2.0 \text{ mm}$. The number of fins that could fit within the allowed width was determined from:

$$W = Nt_f + (N - 1)s \quad (5)$$

Rearranging this equation for N:

$$N = \left\lceil \frac{W + s}{t_f + s} \right\rceil \quad (5.1)$$

$$N = \left\lceil \frac{70.5 + 2.0}{1.0 + 2.0} \right\rceil = 24 \text{ fins}$$

Since the exact geometry of the integrated heat-spreader of the processor is not publicly available, the effective contact area between the heatsink base and the processor was modeled as a fraction of the top area. The final design used a conservative assumption of 95% effective contact. If the package area is denoted by A_{pkg} , the effective contact area is:

$$A_{eff} = 0.95A_{pkg} \quad (6)$$

Using the previously established package area of $A_{pkg} = 4.37875 \cdot 10^{-3} \text{ m}^2$:

$$A_{eff} = 0.95(4.37875 \cdot 10^{-3}) = 4.15981 \cdot 10^{-3} \text{ m}^2$$

Copper was selected for the heatsink design because the dominant internal conduction terms scale inversely with thermal conductivity. Since copper has a thermal conductivity of $k_{Cu} = 390 \text{ W/mK}$, it is substantially more effective as a thermal spreader than aluminum, especially when the heat source is concentrated over a small contact area.

A thin thermal interface layer was assumed to be between the processor and heatsink base held in place by thermal paste. The thermal resistance of this layer was modeled as:

$$R_{\theta,TIM} = \frac{t_{TIM}}{k_{TIM}A_{eff}} \quad (7)$$

Using $t_{TIM} = 6.0 \cdot 10^{-5} \text{ m}$, $k_{TIM} = 5 \text{ W/mK}$ with the value of A_{eff} gives:

$$R_{\theta,TIM} = \frac{6.0 \cdot 10^{-5}}{5(4.15981 \cdot 10^{-3})} = 0.00288 \text{ K/W}$$

The conduction resistance through the copper base was modeled as one-dimensional through-thickness conduction by:

$$R_{\theta,b} = \frac{t_b}{k_bA_{eff}} \quad (8)$$

Substituting known values gives:

$$R_{\theta,b} = \frac{0.003}{390(4.15981 \cdot 10^{-3})} = 0.00185 \text{ K/W}$$

The design was treated as a passive heatsink, so the sink-to-air was modeled using natural convection and radiation. The finned surfaces were approximated as vertical surfaces with characteristic length equal to the height of $H = 0.0823 \text{ m}$. The equation for the Rayleigh Number is:

$$Ra_H = \frac{g\beta\Delta TH^3}{\nu\alpha} \quad (9)$$

Equation 10 can be used to calculate β :

$$\beta = \frac{1}{T_f} \quad (10)$$

T_f can be calculated with:

$$T_f = \frac{T_s + T_\infty}{2} \quad (11)$$

After converting T_s and T_∞ from °C to K:

$$T_f = \frac{358.15 + 293.15}{2} = 325.65 \text{ K}$$

$$\beta = \frac{1}{325.65} = 0.00307 \text{ K}^{-1}$$

Using $g = 9.81 \text{ m/s}^2$ with approximate air values near the film temperature of $\alpha = 1.8 \cdot 10^{-5} \text{ m}^2/\text{s}$ and $\nu = 2.6 \cdot 10^{-5} \text{ m}^2/\text{s}$ gives:

$$Ra_H = \frac{(9.81)(0.00307)(65)(0.0823)^3}{(2.6 \cdot 10^{-5})(1.8 \cdot 10^{-5})}$$

$$Ra_H = 2.33 \cdot 10^6$$

The Nusselt number for a vertical surface is estimated using:

$$Nu_H = 0.68 + \frac{0.670 Ra_H^{\frac{1}{4}}}{\left[1 + \left(\frac{0.492}{Pr}\right)^{\frac{9}{16}}\right]^{\frac{4}{9}}} \quad (12)$$

Using an approximate air value of $Pr = 0.71$ with the found value for Ra_H gives:

$$Nu_H = 0.68 + \frac{0.670(2.33 \cdot 10^6)^{1/4}}{\left[1 + \left(\frac{0.492}{0.71}\right)^{9/16}\right]^{4/9}}$$

$$Nu_H = 0.68 + \frac{26.177}{1.309}$$

$$Nu_H = 20.78$$

Equation 13 is used to calculate the convection coefficient as follows:

$$h_{conv} = \frac{Nu_H k_{air}}{H} \quad (13)$$

Using an approximate air value of $k_{air} = 0.028 \text{ W/m} \cdot \text{K}$ with the found value for Nu_H gives:

$$h_{conv} = \frac{(20.78)(0.028)}{0.0823}$$

$$h_{conv} = 7.07 \text{ W/m}^2 \cdot \text{K}$$

Radiative heat transfer was incorporated through a linearized radiation coefficient with:

$$h_{rad} = \varepsilon\sigma(T_s + T_\infty)(T_s^2 + T_\infty^2) \quad (14)$$

A high-emissivity black surface finish was assumed, so $\varepsilon = 0.95$. Using the surface and ambient temperatures gives:

$$h_{rad} = (0.95)(5.67 \cdot 10^{-8})(358.15 + 293.15)[(358.15)^2 + (293.15)^2]$$

$$h_{rad} = (0.95)(5.67 \cdot 10^{-8})(358.15 + 293.15)(128271.4225 + 85936.9225)$$

$$h_{rad} = 7.51 \text{ W/m}^2 \cdot \text{K}$$

The combined effective sink-to-air coefficient is, therefore:

$$h_{eff} = h_{conv} + h_{rad} \quad (15)$$

$$h_{eff} = 7.07 + 7.51 = 14.59 \text{ W/m}^2 \cdot \text{K}$$

The exposed surface area for one fin follows:

$$A_{fin} = 2HL + t_f L \quad (16)$$

The exposed surface area for the full array is, therefore:

$$A_{fins} = N(2HL + t_f L) \quad (17)$$

The exposed surface area for the base is:

$$A_b = (N - 1)sL \quad (18)$$

So, the total exposed area is:

$$A_t = A_{fins} + A_b \quad (19)$$

Substituting the chosen and calculated values gives:

$$A_t = N(2HL + t_f L) + (N - 1)sL \quad (19.1)$$

$$A_t = 24[2(0.083)(0.095) + (0.001)(0.095)] + (24 - 1)(0.002)(0.095)$$

$$A_t = 0.38194 \text{ m}^2$$

The sink-to-air resistance is computed with:

$$R_{\theta, sink-air} = \frac{1}{h_{eff} A_t} \quad (20)$$

$$R_{\theta, sink-air} = \frac{1}{(14.58)(0.38194)}$$

$$R_{\theta, sink-air} = 0.1795 \text{ K/W}$$

The total thermal resistance is, therefore:

$$R_{\theta, total} = 0.1795 + 0.00288 + 0.00185$$

$$R_{\theta, total} = 0.18427 \text{ K/W}$$

Equation 1 can be used to then calculate the maximum power dissipated under the given constraints:

$$Q_{max} = \frac{65}{0.18427}$$

$$Q_{max} = 352.7 \text{ W}$$

Thus, the selected geometry is expected to dissipate approximately 353 W.

Figures

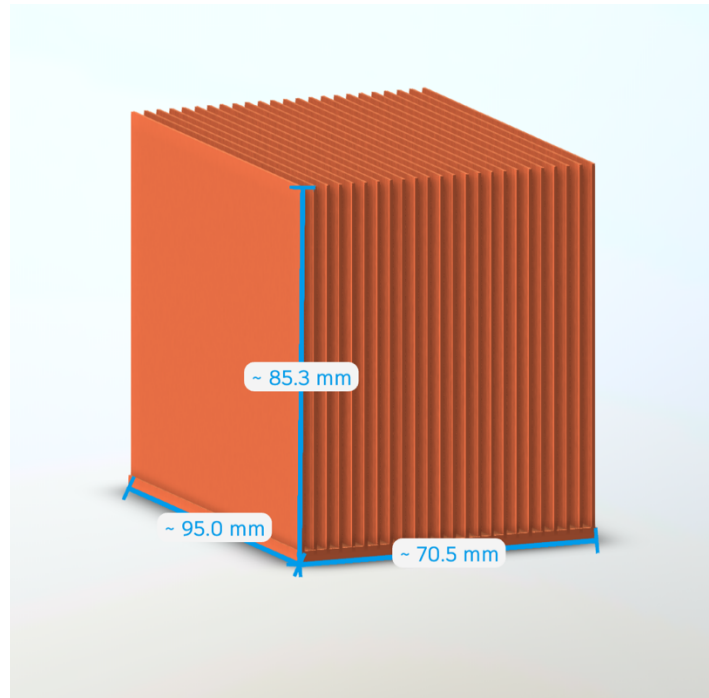


Figure 1: Isometric View of Final Straight-Fin Heatsink Model

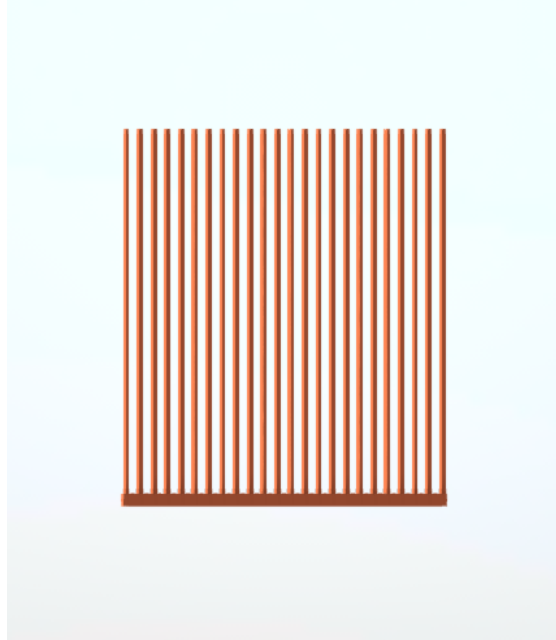


Figure 2: Front View of Final Straight-Fin Heatsink Model



Figure 3: Side View of Final Straight-Fin Heatsink Model

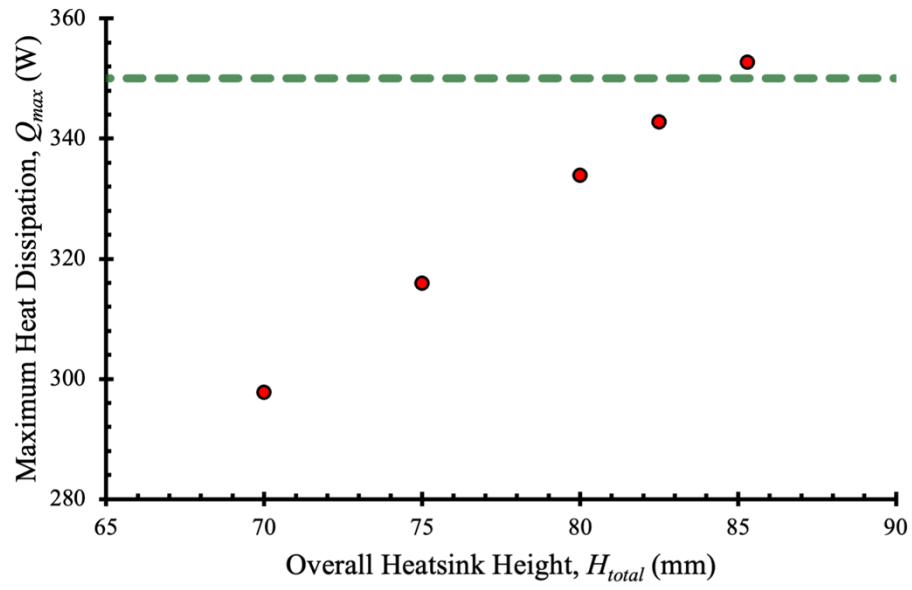


Figure 4: Effect of Overall Heatsink Height on Maximum Heat Dissipation

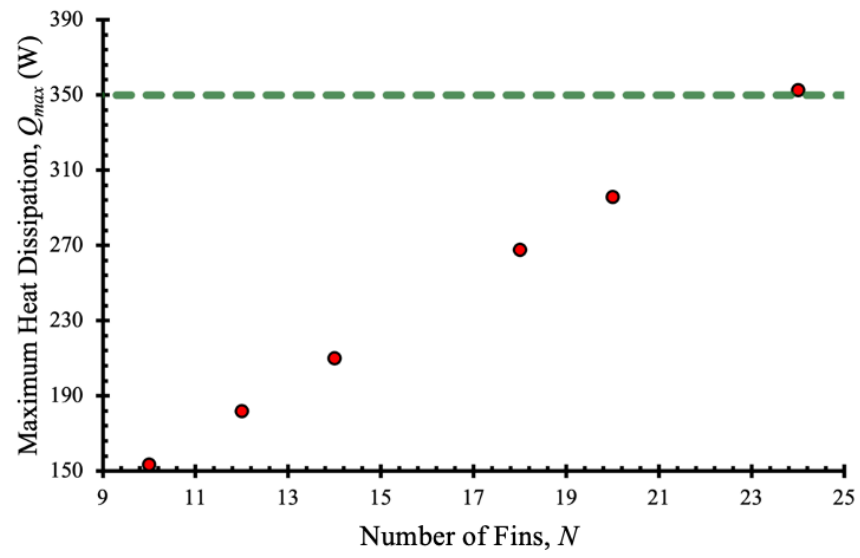


Figure 5: Effect of Fin Count on Maximum Heat Dissipation